

Course Code : ECT 359-3

QTSR/RW – 24 / 1670

**Sixth Semester B. Tech.(Electronics and Communication Engineering)
Examination**

BIOMEDICAL ELECTRONICS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Assume suitable data wherever necessary.
- (3) Illustrate your answers with neat sketches.

1.
 - (a) Describe four major physiological system of the human body mentioning their importance and role. 6(CO1)
 - (b) Explain propagation of action potentials with reference to resting, depolarization and repolarization. 4(CO1)
2.
 - (a) Explain with function diagram of an Electrocardiogram biomedical recorder. 6(CO2)
 - (b) List the type of electrodes for bio-potential measurement. Select and justify the type of electrode from this list for ECG measurement. 4(CO2)
3.
 - (a) Elaborate what do you mean by biotelemetry? Explain the necessary elements of biotelemetry system with block schematic. 6(CO2,4)
 - (b) Discuss EOG and ERG bioelectric signals. 4(CO2,4)
4.
 - (a) List the purpose of ultrasonic imaging for medical diagnosis. Describe the functioning of ultrasonic imaging equipment. 6(CO4)
 - (b) Discuss the precautions to be taken before MRI scanning. 4(CO4)

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5. (a) What is the need of Ambulatory Monitoring Systems? Explain the subsystems for data recording and analysis in an ambulatory monitoring system with an example. 6(CO2,3,4)
- (b) Describe the organization of intensive care operating room. 4(CO4,5)
6. (a) Discuss physiological effects of electrical shock to human body and the precautions to prevent it. 6(CO5)
- (b) Differentiate Micro and Macro shock hazards 4(CO5)

